Biology and Environmental Science (CY202)

Unit 4: Air and Noise Pollution — Question Bank & Study Guide

Source: Unit 4 Question Bank & Class Notes (Dept. of Chemistry)

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Compiled as: Section A – Two Mark Questions | Section B – Seven Mark Questions

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Section A: Two Mark Questions

Short-answer questions — 2 marks each

Topic 1: Basics of Air Pollution

Q1. What do you mean by air pollution?

Air pollution is the release into the atmosphere of gases, finely divided solids, or finely dispersed liquid aerosols at a rate that exceeds the natural capacity of the environment to dissipate, dilute, or absorb them, such that the resulting concentrations cause undesirable health, economic, or aesthetic effects.

Topic 2: Particulate Matter & Trace Pollutants

Q2. Define particulate matter.

Particulate matter (PM) refers to airborne suspensions of extremely small solid or liquid particles — such as soot, dust, smoke, fumes, and mists — especially those under 10 micrometres (µm) in size. PM is significant because it can be inhaled deep into the lungs and cause serious harm to human health.

Q3. Mention the division of particulate matter with dimension.

- **Dust particles:** solids roughly 1–100 µm in diameter.
- **Fumes:** airborne solids less than 1 µm in diameter.
- **Coarse PM:** 2.5–10 µm; **Fine PM:** < 2.5 µm; **Ultrafine PM:** < 0.1 µm.
- Particles under 10 µm are of greatest health concern since they penetrate deep into the lower respiratory system.

Q4. Name the major six air pollutants.

The six "criteria" pollutants designated by the U.S. EPA are: Carbon monoxide (CO), Lead (Pb), Nitrogen dioxide (NO_2), Ozone (O_3), Particulate Matter (PM), and Sulfur dioxide (SO_2).

Q5. What are the trace pollutants?

Trace pollutants are gases present in very small (trace) amounts in otherwise clean, dry air (which is ~78% nitrogen and ~21% oxygen). The remaining ~1% is mostly argon along with trace amounts of carbon dioxide, methane, hydrogen, helium, and others — some of which become significant pollutants at elevated concentrations.

Topic 3: Criteria Air Pollutants (Overview)

Q6. What are the criteria for gaseous pollutants in urban regions?

The gaseous criteria pollutants of primary concern in urban settings are sulfur dioxide, nitrogen dioxide, and carbon monoxide — emitted directly from fossil fuels (fuel oil, gasoline, natural gas) burned in power plants, automobiles, and other combustion sources.

Topic 4: Smog, Fog & VOCs

Q7. What is photochemical smog? How is it different from smog?

Photochemical smog is a reddish-brown haze formed when nitrogen dioxide reacts with volatile organic compounds (VOCs) in the presence of sunlight, producing ozone as a key component. Ordinary "smog" is simply smoke combined with fog (water droplets in air), whereas photochemical smog is produced by sunlight-driven chemical reactions rather than the physical mixing of smoke and fog.

Q8. What are VOCs?

Volatile Organic Compounds (VOCs) are organic compounds that readily evaporate, including pure hydrocarbons, partially oxidized hydrocarbons, and compounds containing chlorine, sulfur, or nitrogen. They are widely used as fuels, paint thinners/solvents, and in plastics production, and contribute to smog formation and greenhouse warming.

Topic 5: Abbreviations

Q9. Abbreviations: NAAQS, EPA, VOC, CFC, CCS etc.

- **NAAQS** — National Ambient Air Quality Standards
- **EPA** — Environmental Protection Agency
- **VOC** — Volatile Organic Compound
- **CFC** — Chlorofluorocarbon
- **CCS** — Carbon Capture and Storage (also Carbon Capture and Sequestration)

Topic 6: Bhopal Gas Tragedy

Q10. What happened during the Bhopal Gas tragedy?

In the Bhopal disaster of 1984, an accidental release of methyl isocyanate gas at a Union Carbide pesticide factory in Bhopal, Madhya Pradesh, India, immediately killed at least 3,000 people, eventually caused an estimated 15,000–25,000 deaths over the following quarter-century, and injured hundreds of thousands more. It remains the most severe example of an "air toxics" release.

Topic 7: Particulate Control Equipment

Q11. What are the optimum sizes of a PM that can be filtered by cyclones, scrubbers, and electrostatic precipitators?

Device	Particle size effectively removed	Typical efficiency
Cyclone	Coarse particles > ~20 µm	~90%
Spray-tower scrubber	Particles > ~8 µm	~90%
Electrostatic precipitator	As small as 1 µm	> 99%
Baghouse filter	~1 µm (some down to 0.01 µm)	~100%

Q12. What are baghouse filters? Mention one merit and demerit of baghouse filters.

A baghouse is an assembly of long, narrow fabric-filter bags suspended in an enclosure; dust-laden air is blown upward through them, particulates are trapped on the fabric, and clean air exits at the top.

- **Merit:** Very high removal efficiency — nearly 100% for particles as small as 1 µm.
- **Demerit:** Fabric filters offer high resistance to airflow, causing substantial energy use, and the air must usually be cooled below 300°C before filtering, adding further energy cost.

Topic 8: Gaseous Pollutant Control & Carbon Capture

Q13. What is carbon sequestration?

Carbon sequestration is the long-term storage of carbon dioxide — either underground in geologic reservoirs (geosequestration) or on Earth's surface in forests and oceans (biological sequestration) — used to reduce atmospheric CO_2 levels and mitigate global warming.

Q14. What are the two basic forms of CCS?

- **Biological CCS:** the natural environment (forests, oceans, wetlands, grasslands) sequesters CO_2 from the atmosphere.
- **Artificial/Geological CCS:** CO_2 emitted from human-made processes is captured and stored in underground geologic formations (e.g., saline aquifers, depleted oil/gas fields).

Topic 9: Noise Pollution — Basics

Q15. Define noise pollution.

Noise pollution is unwanted or excessive sound that has deleterious effects on human health, wildlife, and environmental quality — i.e., the presence of unwanted, undesirable, and unnecessary sounds in the environment.

Q16. What is the normal sound detection range for humans?

Humans can detect sounds in the frequency range of 20 Hz to 20,000 Hz (20 kHz). In terms of loudness, the human ear can detect sound pressure levels from 0 dB (hearing threshold) up to about 140 dB, with 120–140 dB causing pain (pain threshold).

Q17. What is the range of man-made noise and environmental noise?

- **Man-made noise** (construction, automobiles, household appliances): ranges from about 30 dB to 140 dB.
- **Environmental noise** (animal sounds, thunderstorms, cyclones, hurricanes): can go up to about 140 dB.

Q18. Write four harmful effects of noise pollution.

1. Hearing problems — reduces hearing capability of the ears.
2. Physical issues — headaches, high blood pressure, heart attacks.
3. Psychological issues — hypertension, depression, anxiety, fatigue, stress.
4. Sleeping disorders and loss of concentration, affecting work/study; wild animals are also disturbed and may flee, faint, or die.

Topic 10: Noise Pollution — Numericals

Q19. Determine the sound pressure level if the rms pressure value is given as 30 µPa.

> $L_p = 20 \log_{10} (p_{rms} / p_{ref})$, where $p_{ref} = 20$ µPa

$L_p = 20 \log_{10} (30/20) = 20 \times 0.176 =$ **3.52 dB**

Q20. Calculate the sound pressure level in decibels for a given sound pressure of 20 × 10³ Micro-Pascal.

Given: $p_{rms} = 20 \times 10^3$ µPa = 0.02 Pa; $p_{ref} = 20$ µPa = 20×10^{-6} Pa

> $L_p = 20 \log_{10} (p_{rms} / p_{ref}) = 20 \log_{10} (0.02 / 20{\times}10^{-6}) = 20 \log_{10}(10^3) = 20 \times 3 =$ **60 dB**

Section B: Seven Mark Questions

Detailed answers — 7 marks each

Topic 1: Criteria Air Pollutants — Sources, Characteristics & Effects

Q1. Discuss the characteristics, primary source, and harmful effects of the criterion air pollutants. (7)

The U.S. EPA designates six "criteria" pollutants whose atmospheric concentrations serve as indicators of overall air quality:

Pollutant	Characteristics	Primary Source	Harmful Effects
Carbon monoxide (CO)	Odourless, invisible gas formed by incomplete combustion; most abundant criteria pollutant	Gasoline-powered highway vehicles (primary); residential heating and industrial processes	Displaces oxygen in the bloodstream → asphyxiation at high concentrations/exposure
Sulfur dioxide (SO_2)	Colourless gas with sharp, choking odour	Combustion of coal/oil containing sulfur, mainly power-generating plants	Eye/throat irritation, lung damage; reacts with O_2 and water vapour to form sulfuric acid → acid rain
Nitrogen dioxide (NO_2)	Pungent, irritating gas	High-temperature combustion — coal power plants and gasoline engines	Pulmonary edema; forms nitric acid (acid rain) and drives photochemical smog formation
Ozone (O_3)	Key component of photochemical smog (tropospheric); criteria pollutant only in the troposphere	Formed via reaction of NO_2 and hydrocarbons in sunlight — common where traffic and sunshine are heavy	Respiratory irritation; damages crops and materials
Particulate matter (PM)	Airborne solid/liquid particles (soot, dust, smoke, fumes, mists), especially < 10 µm	Power plants, industrial processes, residential heating, vehicles	Inhaled deep into lungs; some (asbestos, soot) are carcinogenic
Lead (Pb)	Airborne particulates < 0.5 µm, very toxic	Oil refining, smelting, industrial activity (historically leaded gasoline)	Learning disabilities, seizures, even death, particularly in children

CoN SoL

PM

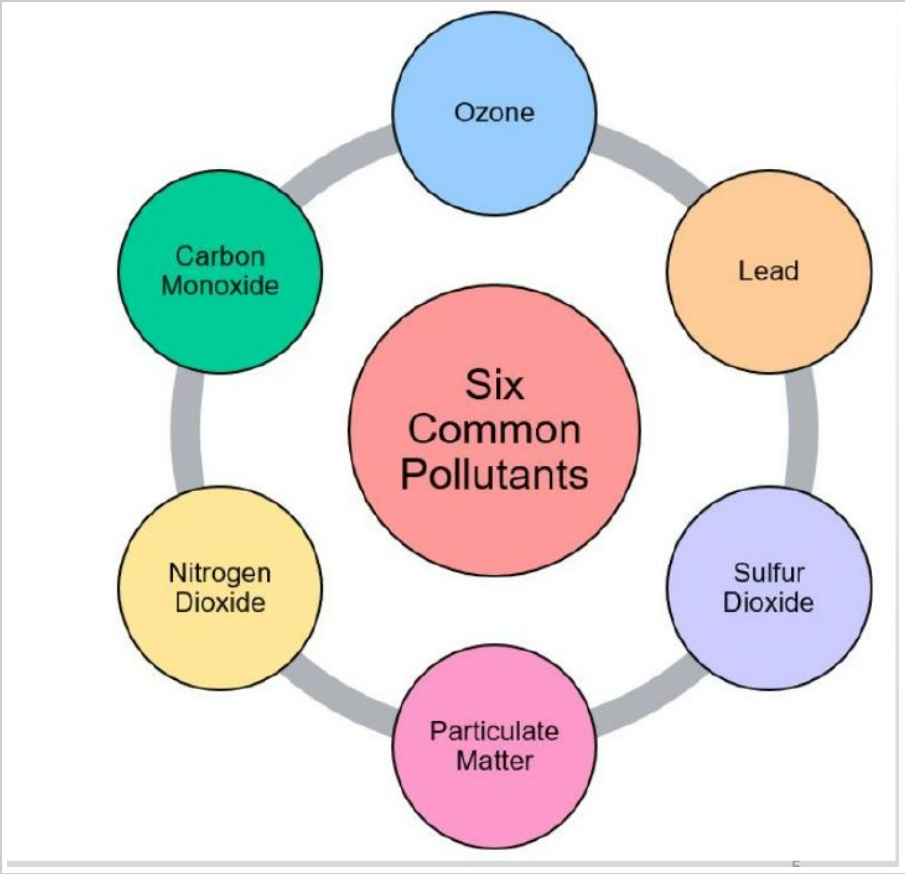


Fig. 1 — The six common (criteria) air pollutants designated by the EPA under the Clean Air Act / NAAQS.

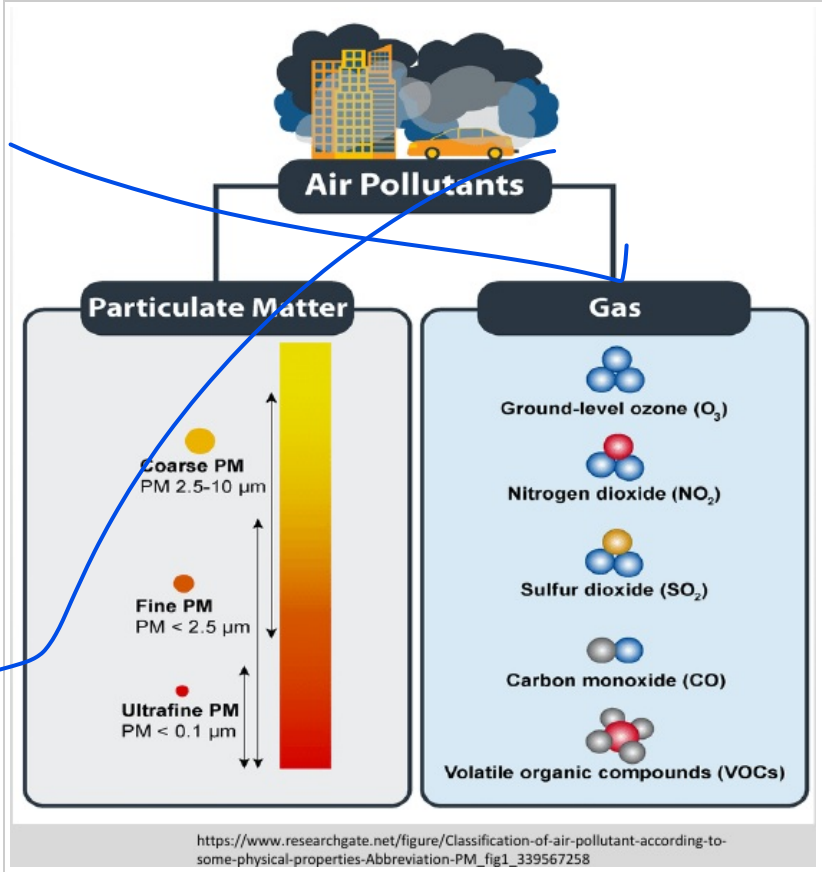


Fig. 2 — Classification of air pollutants into particulate matter (coarse/fine/ultrafine PM) and gaseous pollutants.

Topic 2: Acid Rain

Q2. What is acid rain? Discuss with reactions, the harmful effects of acid rain. (7)

Acid rain is precipitation that has become abnormally acidic due to the reaction of atmospheric sulfur dioxide (SO_2) and nitrogen dioxide (NO_2) — emitted mainly from power plants and vehicles — with oxygen and water vapour in the atmosphere.

Formation reactions:

$SO_2 + O_2 + H_2O \rightarrow H_2SO_4$ (sulfuric acid)
$NO_2 + H_2O \rightarrow HNO_3$ (nitric acid)

These acids fall to the ground as "acid rain," which can also deposit as dry acidic particles.

Harmful effects:

- Harms or destroys fish and plant life in lakes and forests.
- Corrodes metals and deteriorates exposed surfaces of buildings and monuments (e.g., the Taj Mahal).
- Kills trees and degrades soil quality.
- Contributes to respiratory problems via associated SO_2/NO_2 exposure.

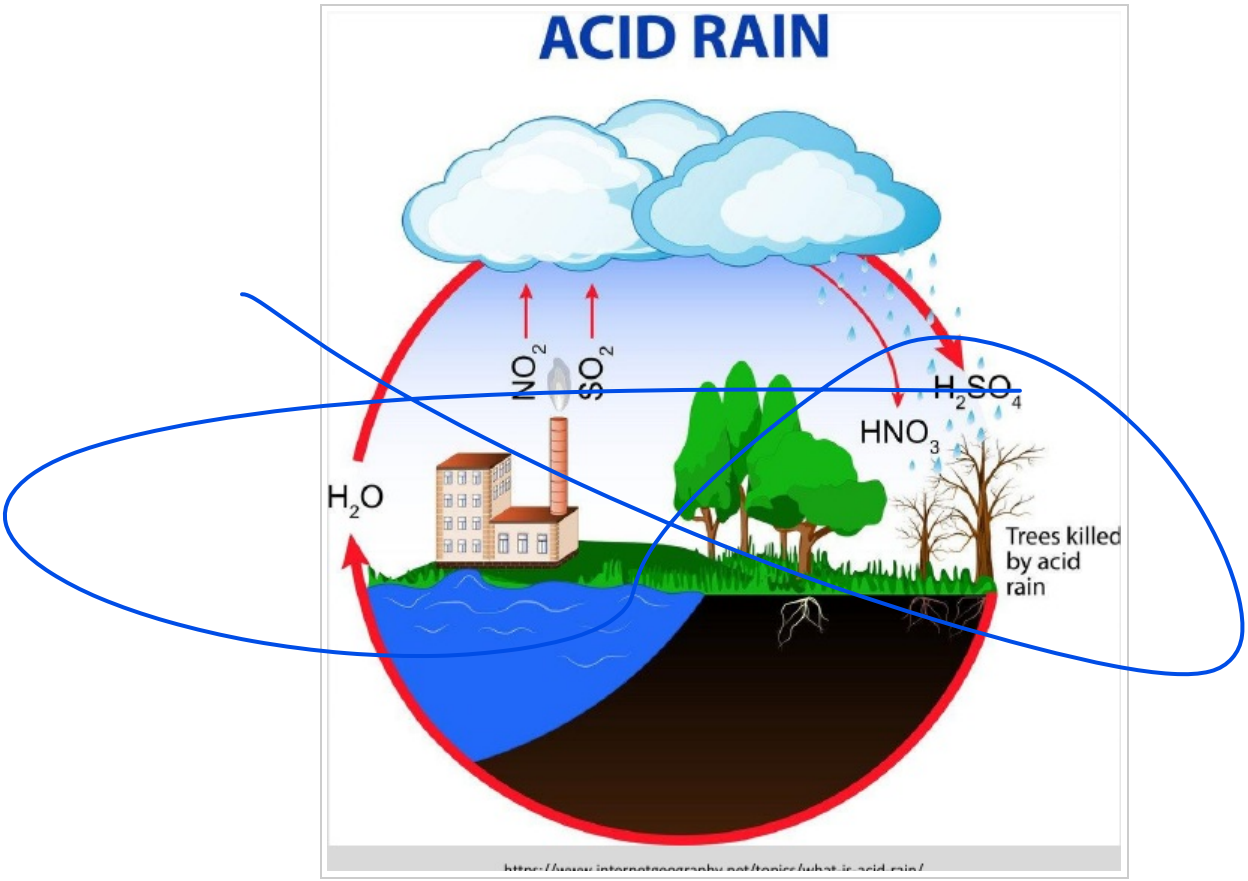


Fig. 3 — Formation cycle of acid rain: SO_2 and NO_2 react with atmospheric moisture to form H_2SO_4/HNO_3, which fall as acid rain and kill vegetation.

Topic 3: Greenhouse Effect, Global Warming & Greenhouse Gases

Q3. What is greenhouse effect? How do greenhouses work? Give examples of four major greenhouse gases. Discuss the adverse effect of the greenhouse effect. (7)

Greenhouse effect: Certain trace gases in the atmosphere trap outgoing infrared (heat) radiation from the Earth's surface, warming the lower atmosphere — analogous to how a glass greenhouse traps heat, letting sunlight in but limiting the escape of heat.

Four major greenhouse gases:

1. Carbon dioxide (CO_2) — most important due to huge volumes emitted from fossil-fuel combustion
2. Methane (CH_4)
3. Nitrous oxide (N_2O)
4. Chlorofluorocarbons (CFCs) and ozone (O_3)

Adverse effects of the greenhouse effect (global warming):

- Rising average global temperatures
- Melting polar ice and glaciers, leading to sea-level rise
- Changing precipitation patterns, more extreme weather events
- Disruption of ecosystems and agriculture

Topic 4: Global Warming — Impact of Greenhouse Gases

Q4. What is global warming? Discuss the impact of greenhouse gases on global warming. (7)

Global warming is the significant environmental problem — recognized by almost all atmospheric scientists — caused by an increase in the levels of certain trace ("greenhouse") gases in the atmosphere since the beginning of the Industrial Revolution (mid-18th century).

- Greenhouse gases include CO_2, CFCs, methane, nitrous oxide, ozone, and others.
- CO_2, although not the most potent per molecule, is the most important contributor because of the huge volumes emitted through combustion of fossil fuels (gasoline, oil, coal).
- VOC emissions also act as greenhouse gases, adding to global warming while also contributing to urban smog and air toxicity.
- Increased greenhouse gas concentration enhances heat trapping in the lower atmosphere, raising average global temperature — driving climate change, sea-level rise, and ecological disruption.
- Mitigation approaches include using energy more efficiently, shifting to alternative energy sources (nuclear, wind, tidal, solar), and carbon sequestration (natural sinks and engineered/geological storage).

Topic 5: Ozone Layer Depletion & CFCs

Q5. What is ozone layer depletion? How does CFC affect ozone depletion? (7)

Ozone layer depletion is the thinning of the stratospheric (upper-atmosphere) ozone layer, which naturally blocks harmful ultraviolet (UV) radiation from the sun. This is distinct from ground-level (tropospheric) ozone, which is a harmful criteria pollutant and component of photochemical smog.

Role of CFCs:

- Chlorofluorocarbons (CFCs) are stable organic compounds released from refrigerants, aerosols, and industrial processes.
- They drift up into the stratosphere, where UV radiation breaks them down, releasing reactive chlorine atoms.
- These chlorine atoms catalytically destroy ozone (O_3) molecules, converting them to oxygen and depleting the protective ozone layer.
- Reduced stratospheric ozone allows more harmful UV radiation to reach the Earth's surface, increasing risks of skin cancer, cataracts, and harm to ecosystems.
- CFCs are also potent greenhouse gases, further linking ozone depletion to global warming.

Topic 6: Electrostatic Precipitator — Working Principle & Efficiency

Q6. Discuss the operational principle of electrostatic precipitator. Comment on its efficiency. (7)

Operating principle:

1. Particles suspended in the polluted airstream are given an electric charge as they enter the unit.
2. A high DC voltage (as much as 100,000 volts) is applied to discharge electrode wires, which hang between rows of large, grounded (positively charged) collection plates arranged vertically and in parallel inside a boxlike structure.
3. The negatively charged wires create a corona discharge that charges the dust particles, which are then attracted to and trapped on the oppositely (positively) charged collection plates.
4. Collected particles form a dust layer (~6 mm thick) on the plates, which is periodically dislodged by mechanically shaking or "rapping" the plates.
5. Dislodged particles fall into a hopper at the bottom of the unit and are removed for disposal.

Efficiency:

- An electrostatic precipitator can remove particulates as small as 1 μm with an efficiency exceeding 99 percent.
- Its effectiveness in removing fly ash from combustion gases of fossil-fuel furnaces accounts for its widespread use at power stations.

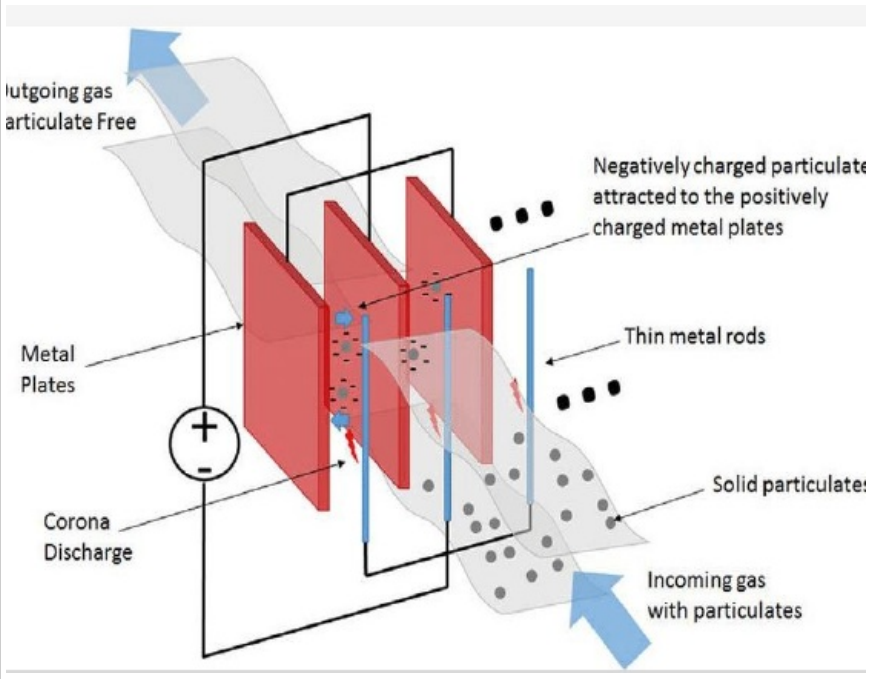


Fig. 4 — Working of an electrostatic precipitator: charged metal plates attract negatively charged particulates via corona discharge, leaving clean outgoing gas.

Topic 7: Flue Gas Desulfurization (FGD)

Q7. Write a short note on flue gas desulfurization. (7)

Flue gas desulfurization (FGD) is an absorption process used to control sulfur dioxide in flue gas from fossil-fuel power plants. FGD systems may involve wet or dry scrubbing.

- **Wet FGD:** flue gases contact a liquid or slurry absorbent; SO_2 dissolves in or reacts with the absorbent and becomes trapped.
- **Dry FGD:** the absorbent is dry pulverized lime or limestone; the resulting solid particles are removed using baghouse filters. Dry systems offer cost/energy savings and easier operation but need higher chemical consumption and work only with low-sulfur coal flue gases.

Key chemical reactions:

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Limestone scrubbing: CaCO₃ + SO₂ → CaSO₃ + CO₂
Lime scrubbing: Ca(OH)₂ + SO₂ → CaSO₃ + H₂O
Oxidation: 2 CaSO₃ + O₂ → 2 CaSO₄ (gypsum)
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- Neither calcium sulfite nor calcium sulfate is very soluble in water; both precipitate out as a slurry by gravity settling.
- The resulting thick slurry, called FGD sludge, creates a significant disposal problem, and millions of tons are generated each year — though FGD does reduce ambient SO_2 and mitigate acid rain.

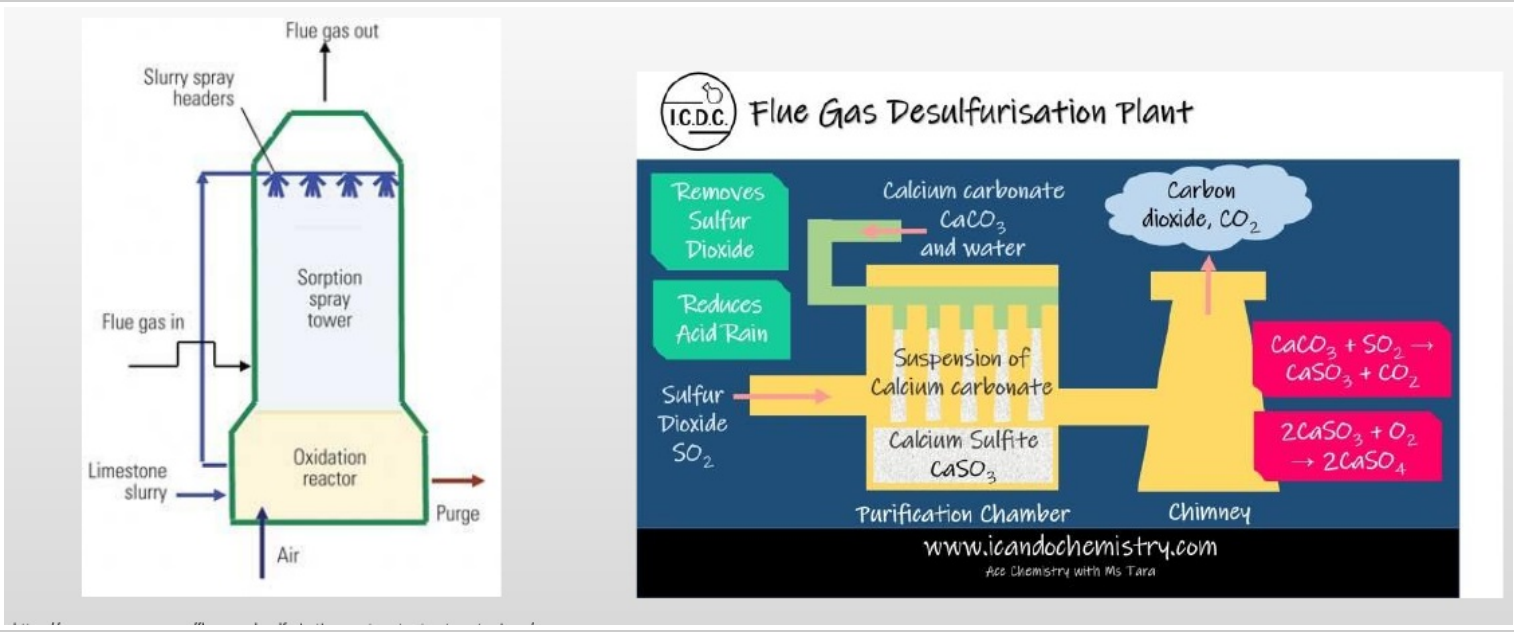


Fig. 5 — Flue gas desulfurization: (left) wet scrubbing sorption spray tower; (right) limestone-based FGD plant removing SO_2 and reducing acid rain.

Topic 8: Noise Pollution — Full Note

Q8. Write a short note on noise pollution. (7)

Definition: Noise pollution is unwanted or excessive sound that has deleterious effects on human health, wildlife, and environmental quality.

Types:

- **Man-made noise:** generated by human activities — construction work, automobiles, household appliances (30 dB–140 dB).
- **Environmental noise:** from natural sources — animal sounds, thunderstorms, cyclones, hurricanes (up to 140 dB).

Sources: bursting of crackers, loud radios/speakers/TV, construction activities, industrial machinery, automobile horns and engines, aircraft (up to 130 dB), home appliances (mixers, washing machines, pressure cookers), animal sounds, and severe weather.

Measurement: Sound waves are vibrations of air molecules; sound cannot travel in a vacuum. Loudness (sound pressure level, SPL) is measured in decibels (dB) using $L_p = 20 \log_{10}(p_{rms}/p_{ref})$, where $p_{ref} = 20$ µPa. Humans can detect 0–140 dB; 120–140 dB causes pain.

Effects: hearing loss, headaches, high blood pressure, heart attacks, hypertension, depression, anxiety, fatigue and stress, sleep disorders, loss of concentration, and disturbance/harm to wildlife.

Prevention/Reduction:

- Avoid playing speakers, radios and television loudly.
- Avoid unnecessary bursting of crackers.
- Avoid unnecessary honking; service vehicles regularly to prevent noisy engines.
- Plant trees around buildings and along roads to buffer noise.
- Separate noise sources (industries, airports) from residential areas.
- Enforce fines for exceeding permissible noise limits and monitor ambient noise levels.

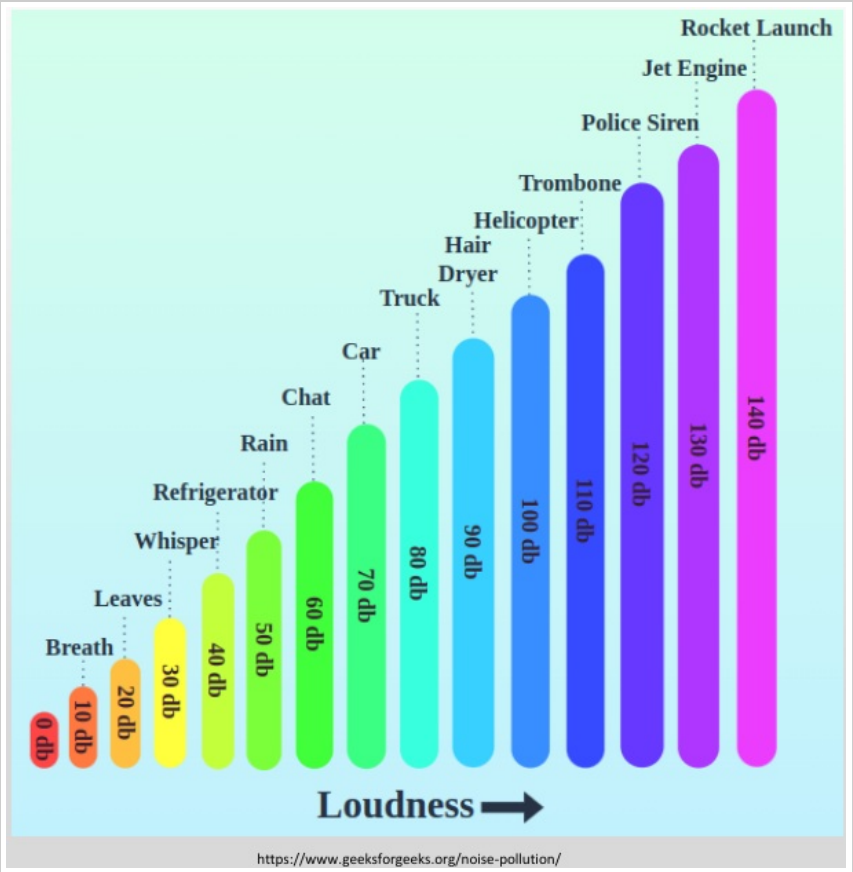


Fig. 6 — Typical loudness (dB) of common sounds, from breathing (0 dB) to a rocket launch (140+ dB).

Topic 9 (Extra): Industrial Air Emission Control Techniques

Q9. Describe the main techniques used to control particulate and gaseous air pollutants at industrial facilities. (7)

(a) Particulate control: Common equipment includes cyclones, scrubbers, electrostatic precipitators, and baghouse filters (see Topic 7 of Section A and Fig. 4 above for ESP detail).

- **Cyclone:** Removes particulates by spinning the dirty airstream in a spiral path inside a cylindrical chamber; larger particles are flung outward by inertia, slide down the wall into a dust hopper, while cleaned air exits through an inner cylinder. Cyclones are best at coarse particles (> 20 µm, ~90% efficiency) and are typically used as pre-cleaners before ESPs/baghouses.
- **Scrubber (wet):** Traps suspended particles by direct contact with a spray of water or liquid; particulates collide with and are entrained by fine droplets. Spray-tower scrubbers remove ~90% of particles > 8 µm.
- **Electrostatic precipitator (ESP):** Charges particles electrically and collects them on oppositely charged plates (see Q6, Section B).
- **Baghouse filter:** Fabric-bag filters trap particulates as air is blown through; nearly 100% efficient for particles as small as 1 µm.

(b) Gaseous pollutant control: Gaseous criteria pollutants, VOCs, and other air toxics are controlled by three basic techniques — used singly or combined — plus a fourth developing technique for CO_2:

1. **Absorption:** Transfers a gaseous pollutant from air into a contacting liquid (e.g., water), which serves as a solvent or reacts chemically with the pollutant. Wet/packed scrubbers achieve 90–95% gas-removal efficiency; used at fertilizer plants (ammonia removal), glass plants (hydrogen fluoride), chemical plants (water-soluble solvents), and rendering plants (odour control). Flue gas desulfurization (FGD) is a major example (see Q7).
2. **Adsorption:** A surface phenomenon where gas molecules are attracted to and held on a solid surface (sorbed). Activated carbon (heated charcoal) is the most common adsorbent, due to its high porosity and surface-area-to-volume ratio; used for odour control, VOC removal, and solvent recovery, achieving > 95% efficiency.
3. **Incineration (combustion):** Converts VOCs and hydrocarbon pollutants to CO_2 and water via rapid oxidation in a special incinerator called an afterburner, which must provide sufficient turbulence, burning time, and temperature. Catalysts (e.g., platinum) allow complete oxidation at lower temperatures. Used at petroleum refineries, paint-drying facilities, and paper mills.
4. **Carbon sequestration:** Long-term storage of CO_2 underground or in natural sinks (see Topic 10 below).



Fig. 7 — A cyclone separator: the swirling gas stream forces coarse particles to the outer wall and down into a collection hopper.



Fig. 8 — A baghouse: dirty air is blown through fabric filter bags; particulates are trapped while clean air exits at the top.

Topic 10 (Extra): Carbon Sequestration & Carbon Credits

Q10. Explain carbon sequestration and how carbon credits help monetise it, as a policy tool for reducing air pollution. (7)

- The best way to reduce atmospheric CO_2 is to use energy more efficiently and shift to alternative energy sources (nuclear, wind, tidal, solar); carbon sequestration serves as an additional tool.
- **Geosequestration (carbon capture and storage):** CO_2 is separated from power-plant flue gases (a costly process) and pumped into underground geologic "reservoir" layers such as depleted oil/gas fields or deep saline aquifers, where it can remain stored for potentially thousands of years.
- **Biological sequestration:** Relies on natural carbon sinks — forests, oceans, grasslands, and wetlands — that absorb CO_2 from the atmosphere; deforestation and ocean pollution diminish this natural capacity.
- **Carbon credits:** A certificate representing removal of one metric tonne of CO_2 from the atmosphere. Companies and governments purchase carbon credits to meet climate commitments (becoming "carbon neutral").
- As a policy measure, initiatives like India's Energy Conservation (Amendment) Bill, 2022 aim to develop a domestic carbon market, indirectly benefiting farmers through soil-carbon-linked incentives, since higher soil carbon also improves soil health, water retention, and reduces flood/drought cycles.

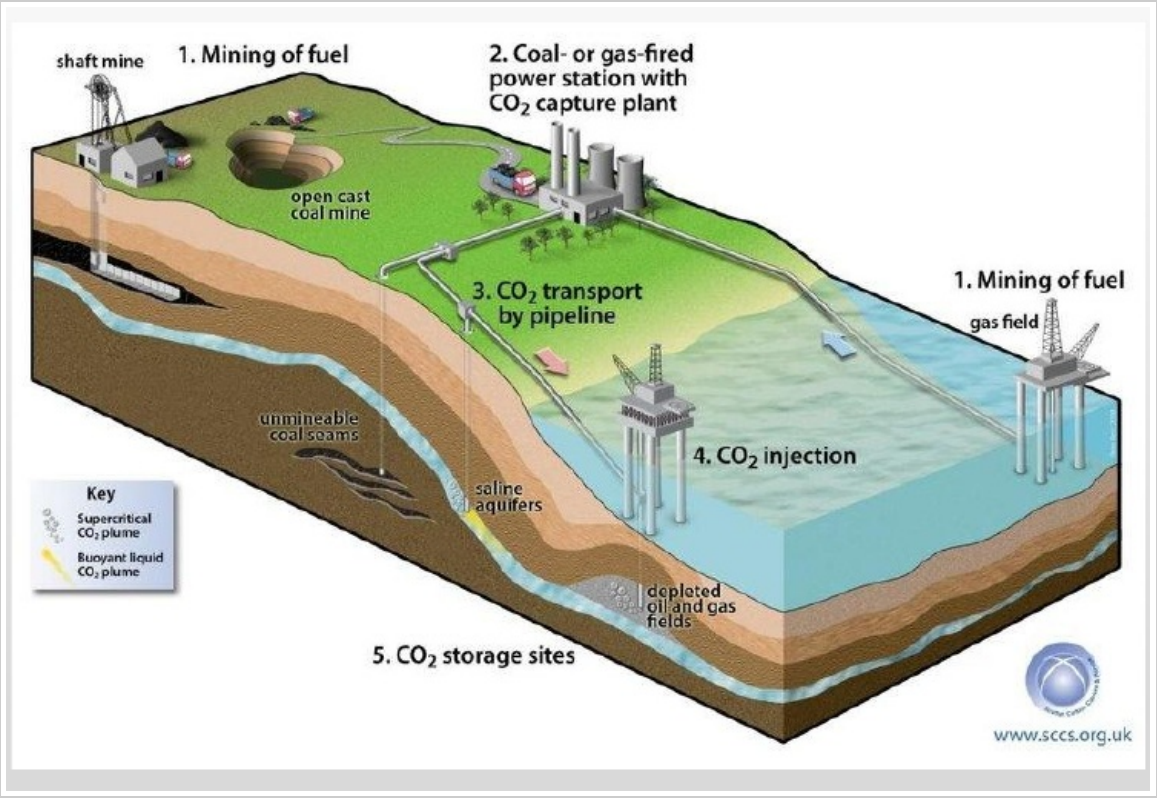


Fig. 9 — Geosequestration: CO_2 captured from a power station is transported by pipeline and injected into underground storage sites (saline aquifers, depleted oil/gas fields).